

Project Title: Parallelized Training of Neural Networks for Biomass Moisture Prediction Using NIR Spectra

Project Goal:

In this project, I aim to build a high-performance system for training artificial neural networks (ANNs) to predict biomass moisture content using near-infrared (NIR) spectroscopy data. The system will use parallel computing to accelerate the training process, enabling faster experimentation with different model architectures and hyperparameters. The main problem of this project which I am trying to solve is that training ANNs on large datasets takes a lot of time, which slows down model development and testing.

Use of Parallel Computing:

Parallel computing is the core part of this project. Training deep learning models takes a lot of computer memory and power, especially when the data is large. To address this problem, I will leverage data parallelism by distributing the training data across multiple CPU cores or GPU threads which make the process very faster. This means that I will split the training data into parts and send each part to a different CPU core or GPU thread. In addition, all parts will be processed at the same time, which will reduce training time of data processing. Additionally, I will explore task parallelism for hyperparameter tuning, where multiple model configurations are trained in parallel. To implement this project, I plan to use technologies such as PyTorch with CUDA or Python's multiprocessing.

Expected Outcome:

I expect to achieve a scalable ANN training pipeline that is significantly faster than the other machine learning models. In this project, performance gains of the machine learning models will be measured based on training time reduction and efficiency of speedup, parallel efficiency etc. The trained models will also be evaluated based on prediction accuracy such as R^2 and RMSE. Overall, the outcome will demonstrate the impact of parallel computing on accelerating machine learning workflows in spectroscopy-based biomass analysis.